

4. A rectangular field is of length 60 m and breadth 22 m. Find the area of the field.

Ans
length = 60 m
breadth = 22 m
area = 1320 m^2

~~formula!~~

5. Arrange the following substances in order of their increasing density
iron, cork, brass, water, mercury

ans cork, water, iron, brass, mercury

ans 6

i Volume of lead piece = 10 cm^3

ii density = $\frac{m}{V}$

3

$$= \frac{88}{10} = 8.8 \text{ gm}^{-3}$$

Motion

i) Keywords:

1. Translatory
2. Rectilinear
3. Curvilinear
4. Rotatory ✓
5. Circular
6. Oscillatory
7. Vibratory ✓
8. Periodic
9. Non-periodic ✓

ii) Important points Formulae

$$1. \text{ Speed} = \frac{\text{Distance travelled}}{\text{Time taken}} \quad \checkmark$$

$$2. \text{ Average speed} = \frac{\text{Total distance travelled}}{\text{Total Time taken}} \quad \checkmark$$

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3. $\text{Weight} = m \times g$
where $m = \text{mass of the object}$
 $g = 10 \text{ N/kg}$

Ans

iii Define the following

1. Rectilinear motion - Motion is a straight line

1

4

2. Curvilinear motion - Motion in a curved path

Ans

3. Uniform motion - Motion in which a body travels equal distances in equal interval of time.

iv Answer the following :

Q1. Explain the meaning of terms rest and motion

Ans Rest : No change in position with respect to ~~motion~~ surroundings.

Motion : The change in position with respect to the surroundings.

1) SAT

4) What do you mean by translatory motion? Give one example

Ans If an object moves in a line in such a way that every point of the object moves through the same distance in the same time is called translatory motions.

ex - The motion of an apple falling from a tree, the motion of a man walking on a road and the motion of a box when pushed from one corner of a room to the other.

Q) What is rotatory motion? Give an example.

Ans A body is said to be in a rotatory motion if it moves about a fixed axis.

Ex - the motion of a ceiling fan

Assignment

3. A boy travels with an average speed of 10 m s^{-1} for 20 min. How much distance does he travel?

Ans Speed = 10 m s^{-1}
Time = 20 min

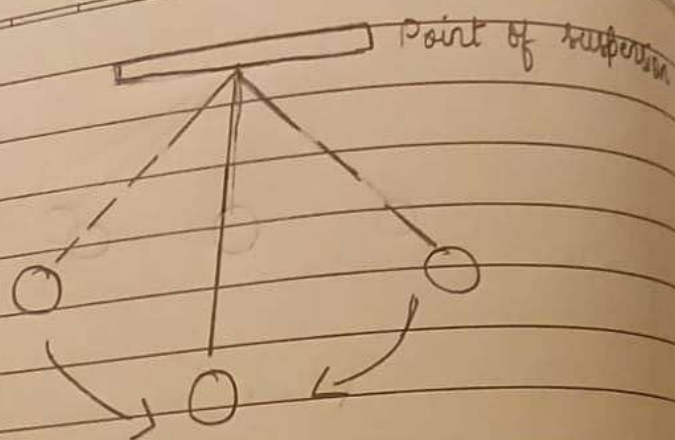
1) How is vibratory motion different from oscillatory motion?

Ans. The vibratory motion, like oscillatory motion is also a two and fro motion but here a part of the body always remains fixed and other parts move two and fro about its mean position.

2) Why is the motion of a flying kite is called random or zig-zag motion?

Ans. It is because the motion of the flying kite does not have a specific path, the direction of changes suddenly.

3) Using an appropriate diagram explain the meaning of oscillatory motion why is it considered a periodic motion?



The to-and-fro motion of an object about a fixed position, as observed in a pendulum or a swing is called oscillatory motion. It is considered as a periodic motion because it repeats itself at regular intervals of time.

Structure Revision

Q1. ~~the SI unit of area~~ Kg per m^2

Q2. 100 m^2

Q3. 2 g/cm^2

Q4. Graph sheet

Q5. Assertion is false but reason is true

ii)

a) speed

b) volume

c) different

iii)

d) length

iv)

d) $\text{Density} = \frac{\text{Mass}}{\text{Volume}} = \frac{252}{30} = 8.4 \text{ g cm cube.}$

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iv mass per unit volume is called density

vi 2100

vii 10^4

viii 2000 m^2

Q3

5. Difference between the following:

a) Rotatory and circular motion.

<u>Rotatory</u>	<u>Circular</u>
- In rotation motion, the axis of the rotation passes from a point in the body itself. - Motion of the earth about its own axis is a rotation motion.	- In circular motion, the axis of revolution passes through a point outside the body. - Motion of satellite around the earth is a circular motion.

b) Periodic and non-periodic motion

<u>Periodic</u>	<u>Non-periodic motion</u>
A motion which gets repeated after a regular interval of time is called a periodic motion.	The motion which does not repeat itself a regular after a regular interval of time is called non-periodic motion.

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Example - Pendulum of a wall clock repeats its motion after every 2 seconds.

Example - A footballer running on a field

c) Mass and weight

Mass

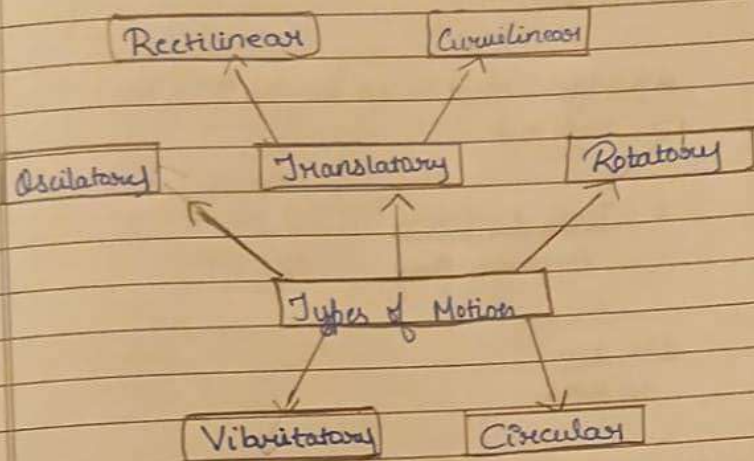
- Mass is the quantity of matter contained in a body
- It is measured by a beam balance
- Its S.I unit is Kilogram (Kg)
- Can never be zero for an object

Weight

- Weight is the force with which the earth attracts the body.
- It is measured by using spring balance
- Its S.I unit is newton (N)
- It can be zero

vi

Concept map:



T.B NUMERICALS Pg no: 36

① $D = 160 \text{ km}$

$t = 4 \text{ hr}$

$$\begin{aligned} \text{Average speed} &= \frac{\text{Total Distance}}{\text{Total Time}} \\ &= \frac{160 \text{ km}}{4 \text{ hr}} \\ &= 40 \text{ km/hr} \end{aligned}$$



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④ Non-uniform motion

$$\text{Average speed} = \frac{\text{Total distance}}{\text{Total Time}}$$

$$= \frac{30\text{ m} + 30\text{ m}}{150\text{ s}}$$

$$= \frac{60\text{ m}}{150\text{ s}}$$

$$= 0.4\text{ m/s}^{-1}$$

⑤ a) $\text{Average speed} = \frac{\text{Total distance}}{\text{Total Time}}$

$$= \frac{1\text{ km} + 0.5\text{ km} + 0.3\text{ km}}{1\text{ h} + 1\text{ h} + 1\text{ h}}$$

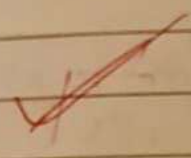
$$= \frac{1.8\text{ km}}{3\text{ h}}$$

$$= 0.6\text{ km/h}$$

(b) $0.6 \text{ km/h} = 0.6 \times \frac{5}{18} \text{ ms}^{-1}$

$= \frac{\cancel{6}^1 \times \cancel{5}^1}{\cancel{10}^2 \cancel{18}^3} = \text{ms}^{-1}$

$= 0.167 \text{ ms}^{-1}$



(6)

③ $m = 37 \text{ kg}$

a) $\text{Weight} = 37 \text{ kgf}$

b) $\text{Weight} = 37 \times 1 \text{ kgf}$
 $= 37 \times 10 \text{ N}$
 $= 370 \text{ N}$

President



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Snap test

i Define the following terms

- a rest
- b rectilinear motion
- c oscillatory motion

ii Differentiate between uniform and non-uniform motion

iii State which of quantities mass/weight is always directed vertically downwards

iv A car covers a distance of 180 km between two cities in 2 hrs. Find its average speed.

v Weight of the object on the earth is 60 N. What is the weight on the surface on the moon

Answers

7/10

1 Rest - a body is said to be at rest if it does not change its position with respect of its fixed point in its surroundings.

2 Rectilinear motion - If the motion of a body is along a straight line.

3 Oscillatory motion - The motion and for motion of a body from its ^{mean} position is called oscillatory motion.

Differentiate between

1 Uniform and non uniform motion

Uniform motion

* It is a moving body travels equal interval of time its motion is said to be uniform motion.

Non uniform motion

* It is moving body travels unequal distances in equal interval of time.



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Light Energy

1. Keywords :

1. Rectilinear propagation
2. Incident ray
3. Point of incidence
4. Reflected ray
5. Normal
6. Angle of incidence
7. Angle of reflection
8. Lateral inversion
9. Regular reflection
10. Irregular reflection

Q Define the following terms:

1. Reflection of light - Bouncing back of light in the same medium when it falls on a surface.

2. primary colours - the colours of light which give white light ~~on the same medium~~ on the mixing together.

3. Angle of Incidence - Angle between the incidence ray and the normal is called angle of incidence.

Q4. Answer the following questions.

1. State the two laws of reflection.

• Angle of incidence is always equal to angle of reflection.

• Incident ray, reflected ray and the normal all lie on the same plane.

2. ~~How~~ How do we see objects around us?

Ans When a ray of light falls on an object it gets reflected. This reflected light reaches our eye & an image of the object is formed on the retina. The information of image & information is sent to the brain and we see the object.

3. A piece of paper appears black in sunlight. What will be its colour when it is seen in red light?

Ans A piece of paper appears black in sunlight because it absorbs the light of all the colours falling on it, hence it will absorb the red light also and will appear black in red light too.